

Basics of Python

1. Which of the following is the correct file extension for Python files?
 - a) .pyth
 - b) .pt
 - c) .py
 - d) .p

Explanation: Python programs are saved with .py extension.

2. Which function is used to display output in Python?
 - a) print()
 - b) show()
 - c) echo()
 - d) display()

Explanation: print() is the standard function to output text.

3. Which keyword is used to define a block of code in Python?
 - a) {}
 - b) begin
 - c) def
 - d) indentation

Explanation: Python uses indentation instead of braces to define blocks.

4. Python is an example of which type of language?
 - a) Compiled
 - b) Interpreted
 - c) Assembly
 - d) Machine

Explanation: Python code runs line by line using an interpreter.

5. What is the correct way to start a comment in Python?
 - a) //
 - b) #
 - c) <!--
 - d) /* */

Explanation: Single-line comments begin with #.

Data Types

6. Which of the following is a mutable data type in Python?
 - a) tuple
 - b) list
 - c) string

d) int

Explanation: Lists can be changed after creation.

7. What is the output of `type(10.0)`?

a) int

b) float

c) str

d) double

Explanation: 10.0 is a floating-point number.

8. Which function is used to find the length of a string?

a) `count()`

b) `len()`

c) `size()`

d) `length()`

Explanation: `len()` gives the total characters in a string.

9. What is the correct data type for True in Python?

a) int

b) str

c) bool

d) float

Explanation: True and False are Boolean values.

10. What is the output of `type("Hello")`?

a) int

b) str

c) list

d) tuple

Explanation: "Hello" is a string.

Operators

11. What is the result of `5 // 2` in Python?

a) 2.5

b) 3

c) 2

d) 2.0

Explanation: `//` gives floor division result.

12. Which operator is used for exponentiation in Python?

a) `^`

b) `**`

c) pow

d) ^^

Explanation: ** raises a number to a power.

13. What will 10 % 3 return?

a) 3

b) 1

c) 0

d) 10

Explanation: % gives the remainder after division.

14. What does the is operator check?

a) Equality of values

b) Identity of objects

c) Membership

d) Bitwise operation

Explanation: is checks if two references point to the same object.

15. What is the result of not (True and False)?

a) True

b) False

c) 0

d) Error

Explanation: True and False is False, not False is True.

Strings

16. Which of the following concatenates two strings?

a) ,

b) +

c) &

d) concat()

Explanation: + joins strings together.

17. What is the output of "Python"[0]?

a) P

b) y

c) n

d) o

Explanation: Indexing starts from 0, so "P" is returned.

18. Which method converts a string to lowercase?

a) lower()

- b) down()
- c) small()
- d) tolower()

Explanation: .lower() converts all characters to lowercase.

19. What is the result of "Hello" * 3?

- a) Error
- b) "HelloHelloHello"
- c) "Hello3"
- d) None

Explanation: * repeats string multiple times.

20. Which escape character is used for a new line?

- a) \n
- b) \t
- c) \r
- d) \b

Explanation: \n inserts a new line.

Lists & Tuples

21. How do you create an empty list in Python?

- a) []
- b) {}
- c) ()
- d) list{}

Explanation: Empty list is [].

22. What is the output of len([1,2,3,4])?

- a) 3
- b) 4
- c) 5
- d) Error

Explanation: The list has 4 elements.

23. Which method adds an element at the end of a list?

- a) insert()
- b) append()
- c) extend()
- d) push()

Explanation: append() adds an item to the list end.

24. Which data type is immutable?

- a) list
- b) dict
- c) tuple
- d) set

Explanation: Tuples cannot be changed once created.

25. What is the output of (1,2,3)[1]?

- a) 1
- b) 2
- c) 3
- d) Error

Explanation: Index 1 refers to 2.

Dictionaries & Sets

26. How is a dictionary defined in Python?

- a) []
- b) ()
- c) {}
- d) <>

Explanation: Dictionaries use {} with key-value pairs.

27. Which method returns all keys of a dictionary?

- a) values()
- b) items()
- c) keys()
- d) allkeys()

Explanation: keys() returns all dictionary keys.

28. What is the output of set([1,2,2,3])?

- a) {1,2,2,3}
- b) {1,2,3}
- c) [1,2,3]
- d) Error

Explanation: Sets remove duplicates automatically.

29. Which operator is used to check membership in sets?

- a) in
- b) has
- c) is

d) exists

Explanation: in checks membership.

30. Which data type does not allow duplicate elements?

a) list

b) tuple

c) set

d) dict

Explanation: Sets store only unique items.

Control Statements

31. Which keyword is used for decision-making in Python?

a) switch

b) if

c) case

d) when

Explanation: if is used for conditions.

32. Which loop is used when number of iterations is unknown?

a) for

b) while

c) do while

d) repeat

Explanation: while runs until condition is false.

33. What does break do in Python?

a) Skips current iteration

b) Stops the loop

c) Pauses the loop

d) Continues the loop

Explanation: break exits the loop completely.

34. What does continue do in Python?

a) Stops loop

b) Restarts loop

c) Skips to next iteration

d) Ends program

Explanation: continue jumps to the next iteration.

35. Which loop is best for iterating over a sequence?

a) for

b) while

c) do while

d) switch

Explanation: for is ideal for sequence iteration.

Functions

36. Which keyword is used to define a function in Python?

a) func

b) def

c) function

d) define

Explanation: def defines a function.

37. What is the default return value of a function if nothing is returned?

a) 0

b) None

c) False

d) ""

Explanation: Functions return None by default.

38. What is the scope of variables defined inside a function?

a) Global

b) Local

c) Public

d) Static

Explanation: Variables inside functions are local.

39. Which keyword is used to return a value from a function?

a) break

b) return

c) yield

d) stop

Explanation: return sends a value back.

40. What is a function inside another function called?

a) Inner function

b) Nested function

c) Both

d) None

Explanation: Function defined inside another is a nested function.

Miscellaneous (Till Functions)

41. What is the output of `print(len("Python"))`?

- a) 5
- b) 6
- c) 7
- d) Error

Explanation: "Python" has 6 letters.

42. What will `range(5)` produce?

- a) [1,2,3,4,5]
- b) [0,1,2,3,4]
- c) [0,1,2,3,4,5]
- d) (1,2,3,4)

Explanation: `range(5)` generates 0 to 4.

43. Which function converts a string to integer?

- a) `str()`
- b) `int()`
- c) `float()`
- d) `chr()`

Explanation: `int("10")` converts string to integer 10.

44. What will be the output of `bool(0)`?

- a) True
- b) False
- c) 0
- d) None

Explanation: 0 is considered False.

45. Which keyword is used to define an anonymous function?

- a) `def`
- b) `lambda`
- c) `anon`
- d) `func`

Explanation: `lambda` creates anonymous functions.

46. What is the output of `sum([1,2,3])`?

- a) 5
- b) 6
- c) [1,2,3]
- d) Error

Explanation: Sum of list elements is 6.

47. Which function gives the maximum value in a list?

- a) `max()`

- b) highest()
- c) maxval()
- d) top()

Explanation: max([1,5,3]) returns 5.

48. Which function is used to get user input in Python 3?

- a) input()
- b) scan()
- c) read()
- d) raw_input()

Explanation: input() is standard in Python 3.

49. Which symbol is used for function arguments with default values?

- a) =
- b) :
- c) ->
- d) ?

Explanation: Default arguments are given using =.

50. What is recursion in Python?

- a) Function calling itself
- b) Function inside class
- c) Function with no arguments
- d) Function with multiple returns

Explanation: Recursion is when a function calls itself.